QM2 Active Learning for Data Collection Optimization

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Background & Motivation:
Quantum materials change behavior rapidly at certain low temperatures; it is desirable to collect data at and around phase transitions.

Current methods of data collection for quantum materials involve measuring data at equi-spaced temperatures rather than at or near transitions, as there is no processing for raw data at the QM2 beamline and processing is a time-intensive process.

By analyzing small amounts of raw data to find when materials start to transition, data can be collected in a more efficient and effective matter, reducing both measurement time, data storage, and processing needs, while also providing more useful data.

Project Approach*:*
Use active learning and Bayseian optimization to analyze raw images to determine phase transitions.

Specifically, using 10 images (taking only 1 second to measure), identify and measure the intensity of regions-of-interest. Then, using Bayesian optimization, predict when material undergoes transition so full data can be collected.

Target Goals, Deliverables, and/or Outcomes
An algorithm that can quickly analyze raw data and determine the next optimal temperature to measure at, made into a functioning pipeline.

Region-of-interest identifier made; next steps are to use Bayesian optimization to find the next temperature to measure, then test the code.